

ARVIND ADITYA MADAN

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Machine learning-focused data scientist with applied experience in predictive marketing, sports analytics, and customer analytics.
Builds models, dashboards, and data workflows that turn messy real-world data into decision-ready insights.

EDUCATION

University of North Carolina at Chapel Hill — *Master of Applied Data Science* **May 2026**

- Relevant coursework: Machine Learning, Deep Learning, Statistical Modeling, Advanced Databases, Visualization & Communication, Governance/Bias/Ethics in Data Science & AI.

University of North Carolina at Chapel Hill — *B.S. Statistics & Analytics* **Dec. 2023**

- Relevant coursework: Sports Analytics, Database Concepts, Optimization, Stochastic Modeling, Linear Algebra, Probability for Data Science, Business Analytics.

TECHNICAL SKILLS

Languages & Tools: Python, SQL, R, Excel, MATLAB, Julia, GitHub

Machine Learning & Data Science: pandas, NumPy, scikit-learn, TensorFlow/Keras, Optuna, predictive modeling, classification, regression, feature engineering, model evaluation

Platforms & Visualization: Databricks, Streamlit, Plotly, Jupyter Notebook, HTML/CSS, D3, dashboard development

PROFESSIONAL EXPERIENCE

ClicMotion — *Junior Data Scientist* **Jun. 2024 – Present**

- Build predictive marketing and customer analytics workflows for large automotive and powersports dealership groups across the United States, using millions of repair order, sales, customer profile, campaign engagement, and third-party household enrichment records.
- Develop service propensity, customer matching, communication preference, and ROI/performance analysis workflows to support sales/service marketing strategy and customer outreach decisions.
- Use Databricks, SQL, Python, and Excel to clean, join, validate, and analyze multi-source dealership/customer datasets across large operational and marketing data environments.
- Support tire, brake, and battery service timing use cases by converting messy dealership data into model-ready features, charts, reports, and business-facing analysis summaries.

SELECTED PROJECTS

UNC Baseball Attendance Forecasting — *MADS Capstone / UNC Athletics*

- Built an end-to-end attendance forecasting framework and Streamlit decision-support app for UNC baseball home games using historical attendance, opponent context, scheduling, weather, academic calendar, Google Trends, promotions, and performance data.
- Led data cleaning, feature engineering, master dataset integration, weighted cross-validation, and preseason workflow adaptation; best two-day-ahead model achieved 260 MAE, 0.8048 R^2 , and 88% within 500 attendees.

MLB Pitch Forecaster — *Machine Learning*

- Developed a multiclass ML pipeline to predict the next pitch type using pre-pitch context, pitch sequencing, pitcher/batter tendencies, handedness, and game state.
- Compared logistic regression, Random Forest, neural network, and ensemble models with Optuna-tuned experiments; final ensemble reached 46% accuracy and 0.42 macro F1 while improving minority pitch-type performance.

ACC Spread Predictor — *Triangle Sports Analytics Competition*

- Predicted second-half ACC men's basketball spreads from first-half-only features using DIFF-based feature engineering, rolling season validation, Random Forest regression, and Optuna tuning; achieved ~9.19 mean MAE and placed 6th in the competition.
- Added split conformal prediction intervals to quantify spread uncertainty while preserving temporal integrity and avoiding leakage across seasons.

CourtSpace 3D — *NBA Shot Visualization*

- Built an interactive Streamlit/Plotly tool for 3D NBA shot visualization with player, season, result, venue, and quarter filters, using NBA API data to explore shot arcs, hot/cold zones, and team/player context.

When Data Bets Against You — *Data Ethics / AI Governance*

- Authored a data ethics analysis of AI-driven sports betting personalization, focusing on behavioral data collection, algorithmic targeting, vulnerable users, responsible gambling systems, and platform governance.

Personal Portfolio Website — *GitHub Pages*

- Designed and deployed a personal portfolio website using HTML/CSS and GitHub Pages to showcase machine learning, sports analytics, data visualization projects, live applications, and full project reports.